

RNA-seq Analysis of SARS-CoV-2 Infection

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2026-03-19

Introduction

RNA sequencing (RNA-seq) enables comprehensive analysis of gene expression across biological conditions (Wang et al., 2009). In this study, differential gene expression was performed to investigate the transcriptional response to SARS-CoV-2 using a publicly available RNA-seq dataset. The aim of the analysis was to identify significant genes that are differentially expressed and to determine their biological pathway in response to viral infection. It was hypothesised that SARS-CoV-2 infection induces differential expression of immune-related and cell cycle-associated genes.

Methods

Setup

```
library(apeglm)
library(ggplot2)
library(ggrepel)
library(pheatmap)
library(DESeq2)
library(enrichplot)
library(clusterProfiler)
library(org.Hs.eg.db)
library(AnnotationDbi)
```

Load Data

```
expr <- read.delim("GSE152418_p20047_Study1_RawCounts.txt", row.names = 1)
```

Metadata

```
samples <- colnames(expr)

meta <- data.frame(sample = samples)
meta$condition <- ifelse(grepl("nCoV|nCOV", samples), "Infected", "Control")
meta$condition <- factor(meta$condition, levels = c("Control", "Infected"))
rownames(meta) <- samples
```

Samples were grouped into infected and control conditions based on dataset annotations, with biological replicates present in each group. Due to lack of explicit metadata annotation, sample conditions were inferred from sample names, which may introduce classification bias

Data Pre-processing

The count matrix was aligned to the sample metadata to ensure correct sample ordering. Lowly expressed genes were filtered out by retaining genes with a minimum total count across samples, reducing noise and improving statistical power for downstream analysis.

```
expr <- expr[, rownames(meta)]
keep <- rowSums(expr) >= 10 # Filter out lowly expressed genes to reduce noise
expr <- expr[keep, ]
```

DESeq2 Analysis

Differential expression analysis was performed using DESeq2, a widely used method for RNA-seq analysis that models count data and controls false discovery rates (Love et al., 2014).

```
# Create DESeq2 dataset with condition as the main variable of interest

dds <- DESeqDataSetFromMatrix(
  countData = expr,
  colData = meta,
  design = ~ condition
)

dds <- DESeq(dds)

# Apply log2 fold change shrinkage to improve effect size estimation

res <- lfcShrink(dds, coef = "condition_Infected_vs_Control", type = "apeglm")
res <- res[!is.na(res$padj), ]
```

The design formula specified condition as the primary explanatory variable, enabling identification of genes associated with infection status. A log2 fold change threshold of 1 was applied to focus on biologically meaningful expression changes, while an adjusted p-value threshold of 0.05 was used to control for false discovery rate.

DESeq2 is widely used for RNA-seq analysis due to its robust modelling of count-based data and control of false discovery rates.

Visualisation

PCA

```
vsd <- vst(dds, blind = FALSE)
pca_plot <- plotPCA(vsd, intgroup = "condition")
pca_plot +
  labs(title = "PCA of Gene Expression")
```

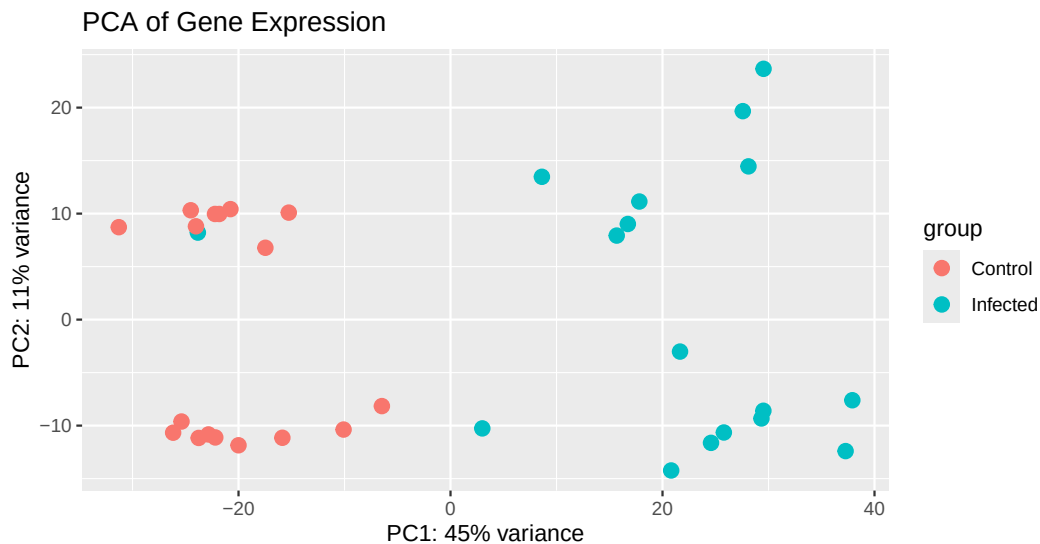


Figure 1: Principal Component Analysis (PCA) showing separation between infected and control samples, indicating that infection status is a major source of variation in gene expression.

Volcano Plot

```
res_df <- as.data.frame(res)

res_df$symbol <- mapIds(
  org.Hs.eg.db,
  keys = gsub("\\\\.*", "", rownames(res_df)),
  column = "SYMBOL",
  keytype = "ENSEMBL",
  multiVals = "first"
)

res_df$group <- "NS"
res_df$group[res_df$padj < 0.05 & res_df$log2FoldChange > 1] <- "Up"
res_df$group[res_df$padj < 0.05 & res_df$log2FoldChange < -1] <- "Down"

top_genes <- head(rownames(res_df[order(res_df$padj), ]), 10)
label_data <- res_df[rownames(res_df) %in% top_genes, ]

ggplot(res_df, aes(x = log2FoldChange, y = -log10(padj), colour = group)) +
  geom_point(alpha = 0.5, size = 1) +

  geom_text_repel(
    data = label_data,
```

```

aes(label = symbol),
size = 3,
max.overlaps = Inf,
box.padding = 0.6,
point.padding = 0.4,
segment.color = "black",
segment.size = 0.3,
force = 2,
show.legend = FALSE
) +

scale_colour_manual(values = c("Down" = "navy", "Up" = "firebrick3", "NS" = "gray")) +
geom_vline(xintercept = c(-1, 1), linetype = "dashed") +
geom_hline(yintercept = -log10(0.05), linetype = "dashed") +
labs(title = "Volcano Plot of Differential Expression") +
theme_minimal()

```

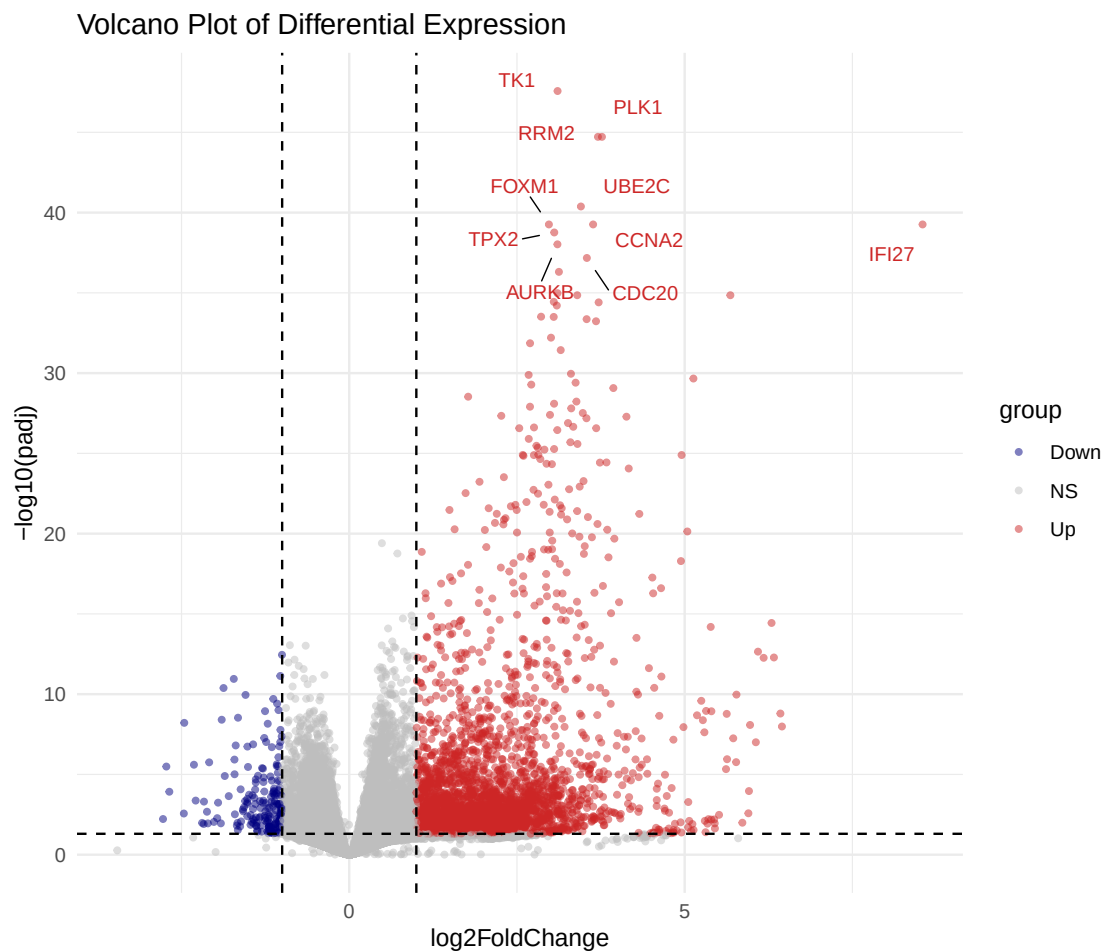


Figure 2: Volcano plot of differential gene expression showing log2 fold change versus statistical significance. Thresholds of $|\log_2\text{FC}| > 1$ and adjusted p-value < 0.05 are indicated by dashed lines. Red points represent upregulated genes, blue points represent downregulated genes, and grey points are not significant.

MA Plot

```
res_df <- as.data.frame(res)

res_df$significant <- "NS"
res_df$significant[res_df$padj < 0.05 & abs(res_df$log2FoldChange) > 1] <- "Significant"

ggplot(res_df, aes(x = baseMean, y = log2FoldChange, colour = significant)) +
  geom_point(alpha = 0.3, size = 0.9) +
  scale_x_log10() +
  scale_colour_manual(values = c("NS" = "grey", "Significant" = "red")) +
  geom_hline(yintercept = 0, colour = "black") +
  geom_hline(yintercept = c(-1, 1), linetype = "dashed") +
  labs(
    title = "MA Plot of Differential Expression",
    x = "Mean of normalized counts",
    y = "Log2 fold change",
    colour = "Significance"
  ) +
  theme_minimal()
```

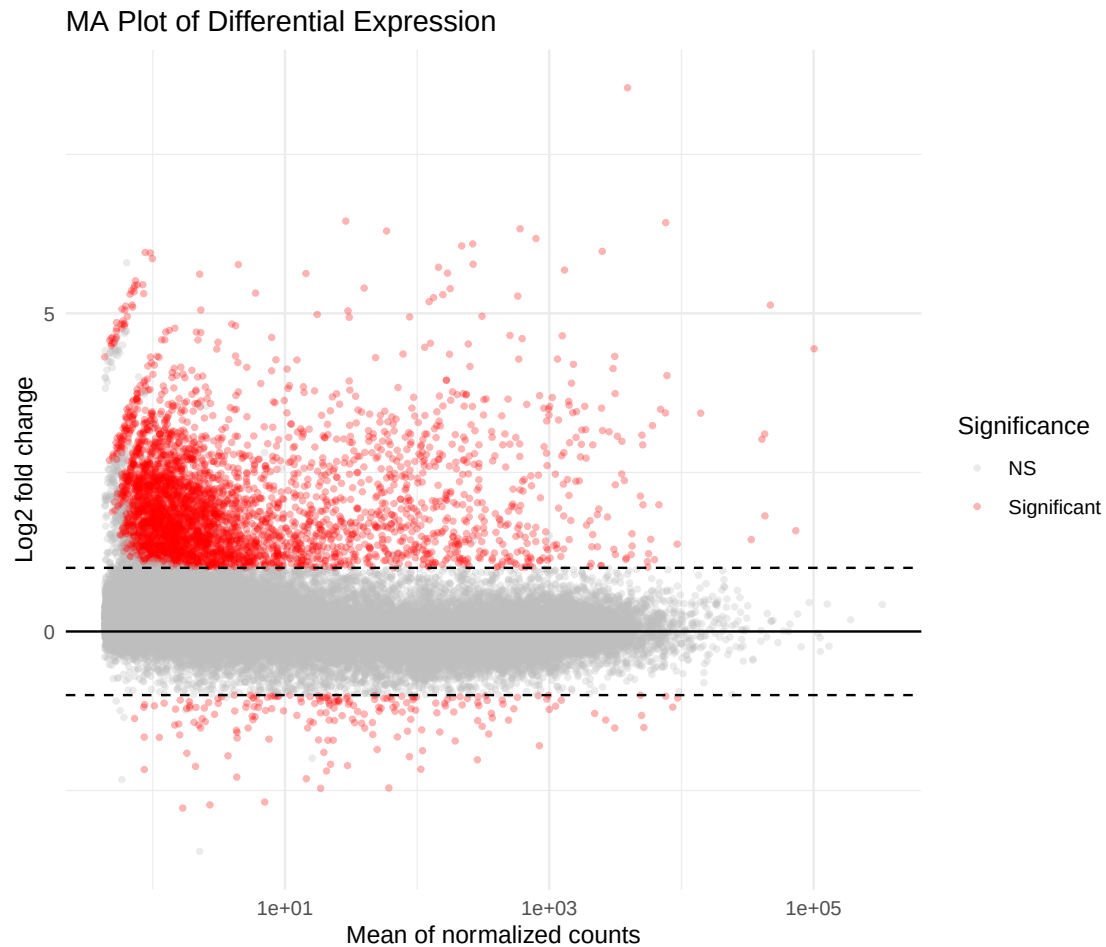


Figure 3: MA plot showing the relationship between mean expression and log2 fold change, demonstrating that differential expression occurs across a wide range of expression levels.

Heat-map

```
top_genes <- rownames(res)[order(res$padj)][1:20]

# Convert ENSEMBL to SYMBOL
gene_symbols <- mapIds(
  org.Hs.eg.db,
  keys = gsub("\\.*", "", top_genes),
  column = "SYMBOL",
  keytype = "ENSEMBL",
  multiVals = "first"
)

heatmap_data <- assay(vsd)[top_genes, ]
rownames(heatmap_data) <- gene_symbols

annotation_col <- data.frame(condition = meta$condition)
rownames(annotation_col) <- rownames(meta)
```

```
pheatmap(
  heatmap_data,
  scale = "row",
  color = colorRampPalette(c("navy", "white", "firebrick3"))(50),
  annotation_col = annotation_col,
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  show_colnames = FALSE,
  fontsize_row = 8
)
```

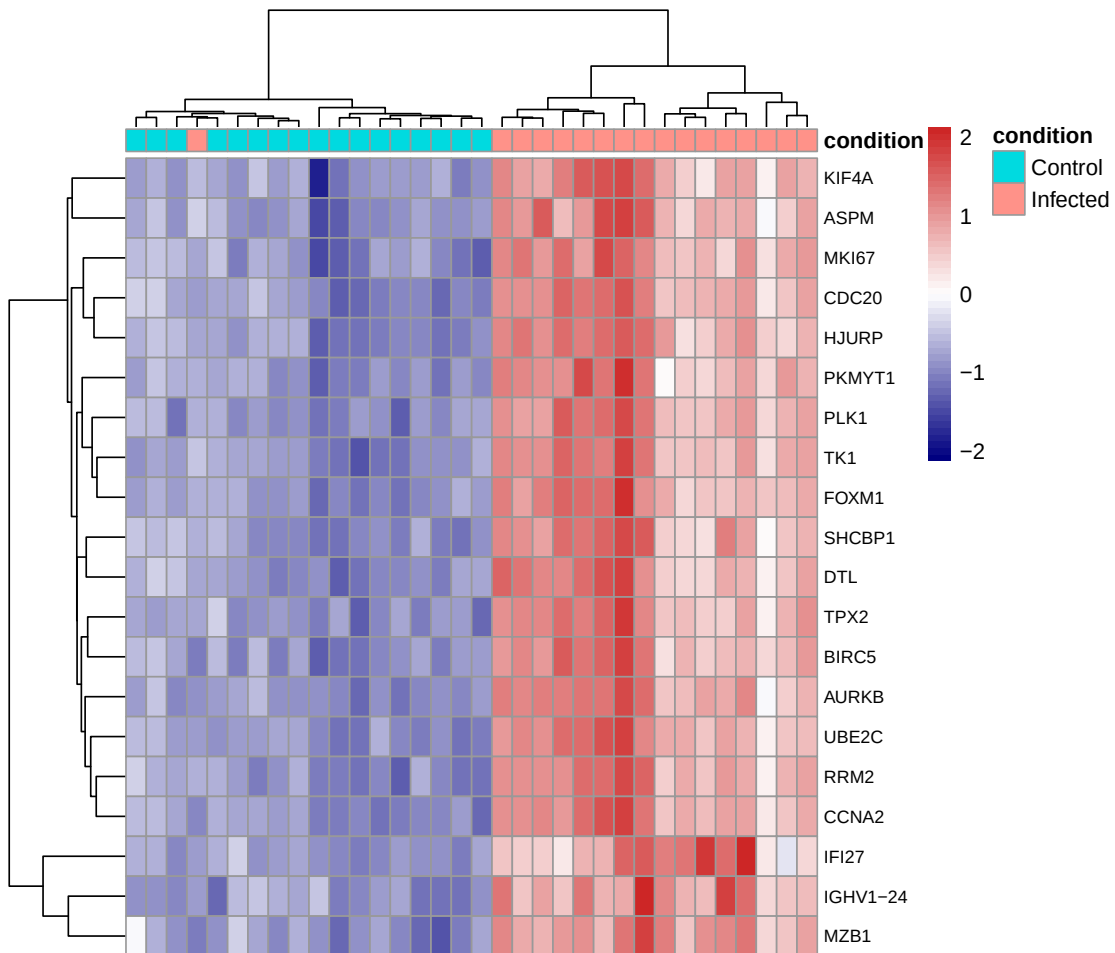


Figure 4: Heatmap of the top 20 differentially expressed genes showing distinct clustering of infected and control samples, indicating consistent expression patterns within each condition.

Sample Distance Heat-map

```
sampleDists <- dist(t(assay(vsd)))

pheatmap(
```

```

as.matrix(sampleDists),
annotation_col = annotation_col,
color = colorRampPalette(c("navy", "white", "firebrick3"))(50),
show_colnames = FALSE,
show_rownames = FALSE,
)

```

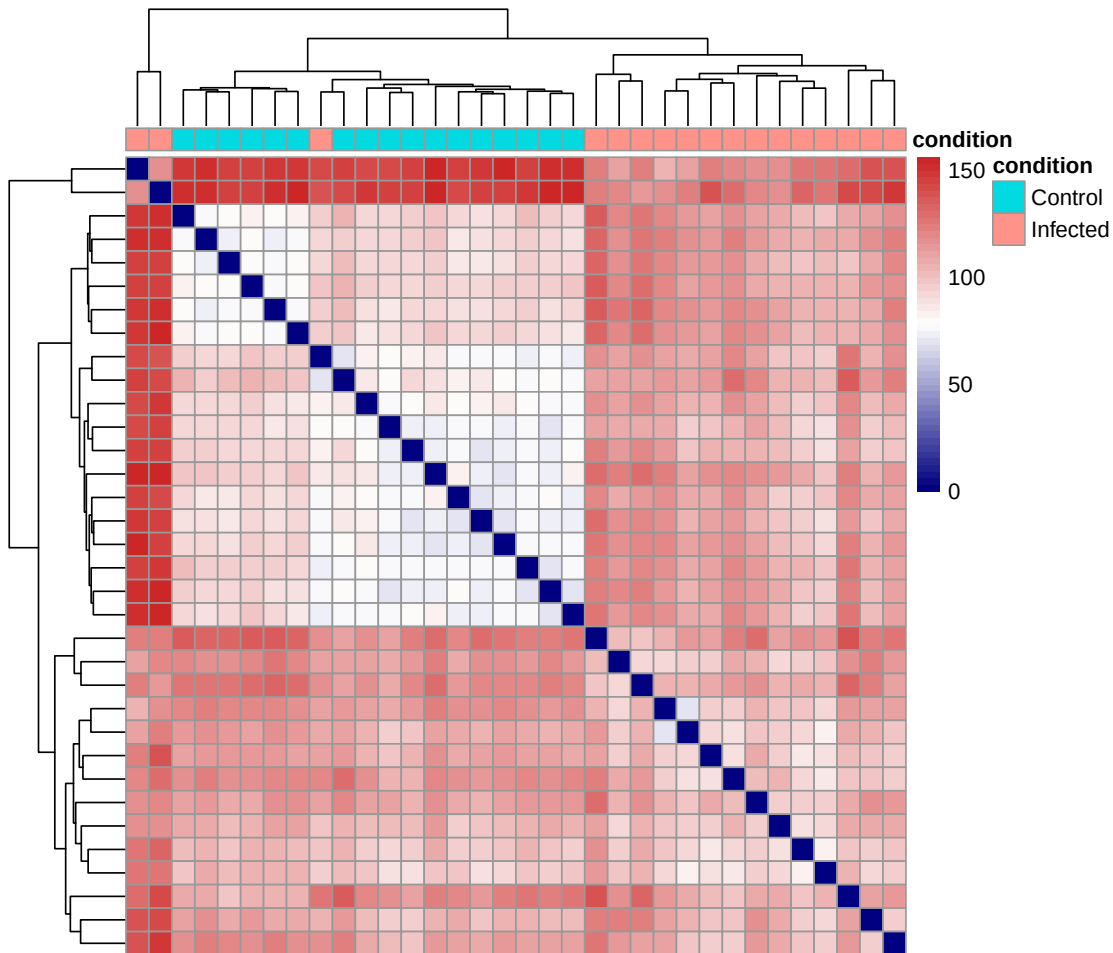


Figure 5: Sample-to-sample distance heatmap demonstrating clustering based on global gene expression profiles, with infected samples grouping separately from controls.

GSEA (GO)

Functional enrichment analysis was conducted using the clusterProfiler package (Yu et al., 2012).

Gene set enrichment analysis was performed to identify coordinated biological pathways associated with SARS-CoV-2 infection (Subramanian et al., 2005).

```

genes <- gsub("\\\\.\\.", "", rownames(res))

gene_list <- res$log2FoldChange
names(gene_list) <- genes

```

```

gene_list <- gene_list[!is.na(gene_list)]

# Convert ENSEMBL IDs to ENTREZ IDs required for GSEA

gene_df <- bitr(
  names(gene_list),
  fromType = "ENSEMBL",
  toType = "ENTREZID",
  OrgDb = org.Hs.eg.db
)

gene_df <- gene_df[!is.na(gene_df$ENTREZID), ]
gene_df <- gene_df[!duplicated(gene_df$ENTREZID), ]

gene_list <- gene_list[gene_df$ENSEMBL]
names(gene_list) <- gene_df$ENTREZID
gene_list <- sort(gene_list, decreasing = TRUE) # Rank genes by log2 fold change for GSEA input

ego_gsea <- gseGO(
  geneList = gene_list,
  OrgDb = org.Hs.eg.db,
  ont = "BP",
  verbose = FALSE
)

dotplot(ego_gsea, showCategory = 8)

```

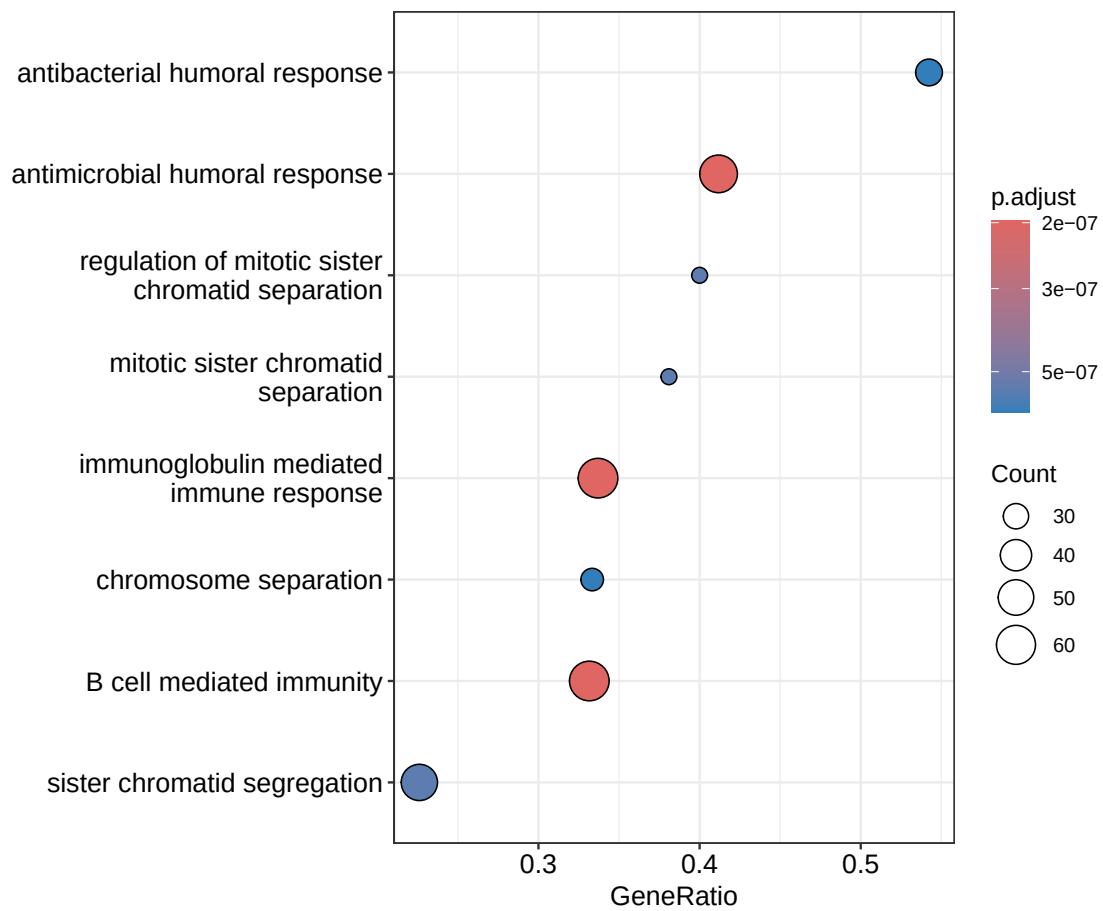


Figure 6: Gene Set Enrichment Analysis (GSEA) dotplot showing significant enrichment of immune-related and cell cycle-associated biological processes.

Enrichment curve (top pathway)

```
gseaplot2(ego_gsea, geneSetID = 1)
```

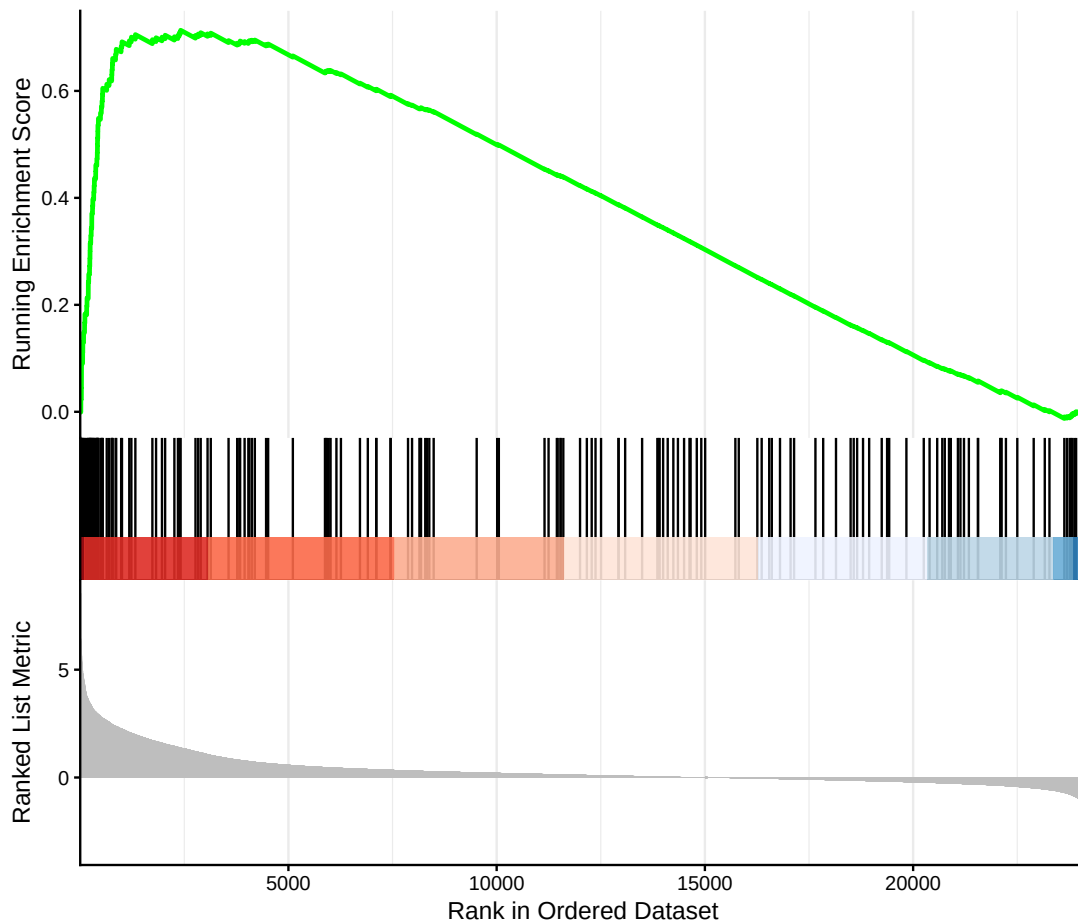


Figure 7: GSEA enrichment plot for the top enriched pathway, illustrating the distribution of pathway-associated genes across the ranked gene list and their contribution to the enrichment score.

Results

Differentially Expressed Genes

```
sig <- res[res$padj < 0.05 & abs(res$log2FoldChange) > 1, ]
```

Differential Expression Summary

```
summary(res, alpha = 0.05)
```

```
##
## out of 33509 with nonzero total read count
## adjusted p-value < 0.05
## LFC > 0 (up)      : 6393, 19%
## LFC < 0 (down)    : 2869, 8.6%
```

```
## outliers [1]      : 0, 0%
## low counts [2]    : 0, 0%
## (mean count < 0)
## [1] see 'cooksCutoff' argument of ?results
## [2] see 'independentFiltering' argument of ?results
```

```
nrow(sig)
```

```
## [1] 3879
```

```
sum(sig$log2FoldChange > 0)
```

```
## [1] 3688
```

```
sum(sig$log2FoldChange < 0)
```

```
## [1] 191
```

Top Differentially Expressed Genes

```
top <- head(sig[order(sig$padj), ], 5)

top$symbol <- mapIds(
  org.Hs.eg.db,
  keys = gsub("\\.*", "", rownames(top)),
  column = "SYMBOL",
  keytype = "ENSEMBL",
  multiVals = "first"
)

top
```

```
## log2 fold change (MAP): condition Infected vs Control
## Wald test p-value: condition Infected vs Control
## DataFrame with 5 rows and 6 columns
##           baseMean log2FoldChange    lfcSE    pvalue    padj
##           <numeric>    <numeric> <numeric> <numeric> <numeric>
## ENSG00000167900    461.825      3.10555  0.205410 7.97098e-53 2.67100e-48
## ENSG00000171848    844.728      3.70698  0.254065 1.71036e-49 1.91042e-45
## ENSG00000166851    185.042      3.76681  0.257566 1.22574e-49 1.91042e-45
## ENSG00000175063    141.186      3.45337  0.248755 4.98148e-45 4.17311e-41
## ENSG00000145386    179.449      3.63535  0.265851 9.18989e-44 5.48343e-40
##           symbol
##           <character>
## ENSG00000167900      TK1
## ENSG00000171848      RRM2
## ENSG00000166851      PLK1
## ENSG00000175063      UBE2C
## ENSG00000145386      CCNA2
```

Enriched Biological Pathways

```
head(ego_gsea@result[, c("Description", "NES", "p.adjust")])
```

```
##                               Description      NES
## GO:0016064      immunoglobulin mediated immune response 1.656567
## GO:0019724                               B cell mediated immunity 1.642341
## GO:0019730      antimicrobial humoral response 1.635432
## GO:0010965 regulation of mitotic sister chromatid separation 1.772262
## GO:0051306      mitotic sister chromatid separation 1.763987
## GO:0000819      sister chromatid segregation 1.485927
##                               p.adjust
## GO:0016064 1.976000e-07
## GO:0019724 1.976000e-07
## GO:0019730 1.976000e-07
## GO:0010965 5.591379e-07
## GO:0051306 5.591379e-07
## GO:0000819 5.591379e-07
```

The results from the differential genes revealed that SARS-CoV-2 infection results in a transcriptional response, where 6393 genes were found to be up-regulated, whereas 2869 were found to be down-regulated (adjusted p-value < 0.05).

Thus, infection status was a major independent source of variation within the dataset, as revealed by PCA, which clearly distinguished between uninfected and infected samples (Figure 1).

Volcano and MA plots show there are a large proportion of up-regulated genes among the significantly differentially expressed genes, suggesting widespread transcriptional activation associated with infection. The volcano plot (Figure 2) shows a strong shift in transcription in favour of upregulation, signalling activated infection-responsive biological pathways. The MA plot also shows that differential expression occurs across a wide range of gene expression levels.

Most importantly, TK1, RRM2, PLK1, UBE2C, and CCNA2 were linked to DNA replication and cell cycle progression. The activation of the genes points to cell cycle associated transcriptional program, and this could also be an indication of immune cell expansion, or a dysregulation of host cells. The heat map of the major differentially expressed genes (Figure 4) shows a grouping together of the infected and controls based on distinct patterns of gene expression under the two conditions. This is further supported by the sample-to-sample distance analysis (Figure 5), with samples from infected populations clustered together and distinguished from controls.

The Gene Set Enrichment Analysis (GSEA) results (Figure 6) indicated significant enrichment of B cell-mediated immunity and immunoglobulin-mediated pathways. The cell-cycle pathways are enriched for immune pathways based on contrasting findings from these studies.

All in all, the increase in immune pathways and upregulation of genes associated with the cell cycle may indicate activation of immune activities. This observation may also suggest an increase in proliferating cells or changes in cell composition prior to actively cycling cells.

Discussion.

The immune pathways were found to be more active due to virus's host response. SARS-CoV-2 infection activates both innate and adaptive immune responses, including B cell activation and antibody production. Some of the enriched pathways include the immunoglobulin-mediated and the humoral immune that point to the transcriptional effect of adaptive immune pathways.

Infection-related immune responses, notably by lymphocytes, may reflect increased expression of cell cycle-associated genes, potentially indicating cellular proliferation or shifts in cell population composition, reflected by expression of cell cycle genes, including PLK1, CCNA2, and RRM2. This may be virus induced cellular stress responses or compensatory mechanisms initiated due to viral replication and host cell damage.

The heat-map and sample distance analysis provided more evidence to back the observations. Also, there was a consistent expression pattern observed in the top DEGs across samples. This shows that the transcriptional reaction to the infection is strong and reproducible.

The results tally with those of other viral infections where activation of the immune response and cellular activation are trademarks for the host. In the case of high prominence of the cell cycle, detailed investigations might be necessary to ensure whether its relevance is due to expansion of immune cells or dysregulation of host's cellular machineries as a consequence of viral infection.

Critical Evaluation

Despite the effectiveness of this analysis, there are several limitations to be taken into account. The use of bulk RNA-seq limits the resolution of cell-type-specific transcriptional responses, making it difficult to determine whether observed changes arise from shifts in cell composition or true transcriptional regulation within individual cell types. This distinction is critical for viral-induced immune activation because there is a chance that the observed effect is due to immune cell infiltration rather than a transcriptional regulation event.

Second, the analysis models infection status as the primary explanatory variable, however, sequencing depth, batch effects, and sample heterogeneity may also contribute to observed variation, and other confounding factors may have an impact on findings and interpretation. Thus, technical confounds may affect the findings and their interpretation, even though the DESeq2 assumes some variation sources' technical nature.

Other than that, an aspect that can affect the identified genes includes the thresholds chosen for significance (adjusted p-value < 0.05 and log2 fold change > 1). Differential enrichment pathways may also be observed following the use of different thresholds.

Also, the use of gene-level summarisation may hide issues with isoform-specific regulation, which may be relevant in viral infection contexts where alternative splicing can occur.

In the end, functional interpretation is left in the hands of annotation databases such as Gene Ontology that can be biased or incomplete to well-characterised genes. It is, therefore, appropriate always to interpret enrichment results cautiously and validate them experimentally when possible.

Future Steps

The results need to be further refined and validated by future studies. It would be a significant area to study based on scRNA-seq's ability to resolve cell-type-specific transcriptional responses. The enrichment of B cell-mediated and immunoglobulin-associated pathways points to the activation of the adaptive immune response, providing evidence that antibody-mediated mechanisms form a key part of the host response to SARS-CoV-2 infection.

Also, if we integrate proteomic or metabolomic data it can also give us an indication of how transcriptional changes manifesting in functional cell responses. Integrating different omics data would enhance biological interpretation of the findings and identify key regulatory mechanisms during infection.

It is also important to validate experimentally the identified differentially expressed genes and pathways. For instance, quantitative PCR (qPCR) or flow cytometry tools could be used to validate gene expression changes and examine immune cell activation in independent samples.

From a computational perspective, another statistical model or machine Learning approaches can also be implemented upon for identifying infection biomarkers & predictive signatures. For instance, the gene expression of control samples and infected samples would be modelled to discriminate the two sample types.

Another analysis that would be helpful is longitudinal or time-course data analyses that understand the timing of the host response to SARS-CoV-2 infection. As a result, identification of early and late response and characterization of disease progression can be done.

Conclusion

In conclusion, there is extensive differential gene expression in response to transcriptional changes triggered by SARS-CoV-2 infection, according to this study. There is also enhanced expression of other cell cycle-related genes with evidence showing a marked activation of immune-related pathways especially B cell and humoral immunity pathways.

The host's immune activation and increased cell cycle activity occur simultaneously to help the host respond against viral infection, the results show.

Collectively, these findings suggest that there is coordinated crosstalk between the activation of the immune system and cell cycle control following viral infection. It reveals circuitry of biological salience that warrants therapeutic exploration in the future

References

- Love, M. I., Huber, W., and Anders, S. (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*, 15(12), 550.
- Wang, Z., Gerstein, M., and Snyder, M. (2009). RNA-Seq: a revolutionary tool for transcriptomics. *Nature Reviews Genetics*, 10(1), 57–63.
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- Yu, G., Wang, L.-G., Han, Y., and He, Q.-Y. (2012). clusterProfiler: An R package for comparing biological themes among gene clusters. *OMICS*, 16(5), 284–287.